Band-1 Hadamard Order 2228 via Miyamoto — Paley-II H_556 + Paley Conference C_558

H_556 Paley-II PASS; H_2228 Miyamoto PASS; SPK gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-14_Band1_Hadamard_2164_Miyamoto`; `2026-09-14_Band1_Hadamard_next_gap_after_2036_`
Lab. `/workspace/experiments/band1/hadamard_2228_miyamoto/`.
Cutoff. 2026-09-14 (~evening ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 556 and 2228 only** (verified $\max|HH^\top - nI| = 0$). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 2228 is a **Miyamoto bonus** (not an SPK fill). `clay_closed = false`.

Abstract

Prior Band-1 ships constructed H_{508}, H_{2004}, H_{2036}, H_{528}, H_{2116}, H_{540}, and H_{2164}. Among remaining Miyamoto-eligible standard-closure misses, this increment constructs **order 2228**: $2228 = 4{\cdot}557$ with 557 prime and $557 \equiv 1 \pmod 4$. On box: H_{556} via Paley type II ($q = 277$, $277 \equiv 1 \pmod 4$), symmetric Paley conference C_{558} over prime 557, and $H_{2228} = \text{Miyamoto}(H_{556}, C_{558})$. All verify $\max |HH^\top - nI| = 0$. Unit tests **17/17 PASS**. SPK next gap remains **2052**; $2228 \notin$ SPK closure (Miyamoto bonus). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 2228; Miyamoto; Paley type II; Paley conference; constructive gap 2052; Band-1; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in $\{\pm 1\}$ satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{556} constructed and verified (Paley type II, $q = 277$). 2. C_{558} constructed and verified (Paley conference, $q = 557$): $CC^\top = 557I$, zero diagonal. 3. H_{2228} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 4. SPK next constructive gap remains **2052** (OPEN). Order 2228 does **not** enter SPK.

2. Arithmetic and eligibility

Quantity	Value
Target order	$2228 = 4 \cdot 557$
$q = 557$	prime, $557 \equiv 1 \pmod 4 \to$ Miyamoto-eligible
Needed H_{q-1}	H_{556}
Needed conference	order $558 = q + 1$
H_{556} path	Paley II: $q' = 277$ prime $\equiv 1 \pmod 4$; order $2(277 + 1) = 556$
Conference path	Symmetric Paley over prime 557

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{556}	Paley type II, $q = 277$	0	0.1141	**PASS**
C_{558}	Paley conference, $q = 557$	0	0.228	**PASS**
H_{2228}	Miyamoto(H_{556}, C_{558})	0	4.9068	**PASS**

Total wall (incl. unit tests + SPK refresh) $\approx$ 5.5128 s. Compressed ± 1 matrices saved as `H556_pm1.npz`, `H2228_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible $\leq 2000 \cup$ on-box $\{508, 2004, 2036, 528, 2116, 540, 2164\} \cup$ verified H_{556} (Paley-II seed) $\cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4000). **Miyamoto is not folded into SPK.**

Quantity	Value
Known count	878
2228 in SPK	**False** (Miyamoto bonus)
2164 in SPK	False (prior Miyamoto bonus)
2116 in SPK	False (prior Miyamoto bonus)
Next standard gap	**2052** (OPEN)
Missing first (sample)	2052, 2060, 2076, 2092, 2108, …

5. Unit tests (17/17)

Test	Result
`H20_paley_I_control`	PASS
`H4_sylvester`	PASS
`paley_II_q5_H12`	PASS
`paley_II_q13_H28`	PASS
`miyamoto_H20_from_H4_conf6`	PASS
`q557_is_prime`	PASS
`q557_equiv_1_mod_4`	PASS
`n2228_equals_4_times_557`	PASS
`q277_is_prime`	PASS
`q277_equiv_1_mod_4`	PASS
`H556_order_formula`	PASS
`C558_order_formula`	PASS
`reject_all_ones_H8`	PASS
`H556_paley_II_eligible`	PASS
`conference_558_shape_expected`	PASS
`conference_6_paley_control`	PASS
`miyamoto_eligible_2228_arith`	PASS

6. Non-claims

- Not the full Hadamard conjecture.
- Not Clay-closed / finished-ToE.
- Not a fill of standard gap **2052**.
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims.
- GATE-04 collection **LOCKED**; no invented J.
- `clay_closed = false`.
- Order 2228 is a **Miyamoto bonus** outside SPK (do not fold Miyamoto into SPK closure).

7. Next optional

Miyamoto **2276** needs H_{568} — Paley-II path with $p = 283 \equiv 3 \pmod 4$ **fails** (needs a different H_{568} construction); OR method-attack standard gap **2052**; OR Track B privacy / Object 1 meter await.

8. Reproducibility

```
cd /workspace/experiments/band1/hadamard_2228_miyamoto
python3 verify_hadamard_2228.py
# emits results.json, SUMMARY.txt, literature_ledger.json,
# H556_pm1.npz, H2228_pm1.npz
```

Artifacts. `results.json`, `SUMMARY.txt`, `literature_ledger.json`, `H556_pm1.npz`, `H2228_pm1.npz`, this paper.